

Chapter 6 Calculate Cost, Environment, and Social Impact Factors

This chapter walks you through all of the R code used to calculate the impact factors that are used in the final model. Additionally, the food group-level inedible and wasted proportions are calculated at the end.

Note that you must first open the 'methods_manual' R project before running this script or else it will not work.

First, let's set up our environment.

```
# check working directory  
getwd()
```

```
## [1] "/Users/bmb73/Library/CloudStorage/Box-Box/lasting_aim_3/model development/methods_n
```



```
# SET UP -----

rm(list = ls())

options(scipen=999)

library(tidyverse)
library(readxl)
library(survey)
library(bookdown)

my_date <- Sys.Date()
```

6.1 Prepare Data

First, we need to import the NHANES dataset to get the **survey weight variables** that are necessary to incorporate when calculating mean estimates from NHANES data. See more information here: <https://wwwn.cdc.gov/nchs/nhanes/tutorials/weighting.aspx>

```
# import subgroup-seqn mapping
nhanes <- read_rds("data_inputs/DIET/dietary_intake/DATA/clean_data/nhanes1518_adj")

# select subset of variables
subgroup_dat <- nhanes %>% select(SEQN, subgroup, SDMVPSU, SDMVSTRA, wtnew, inAnal
```

Then, create a cleaning function that will be used to **clean** the output summary data from the “svymean” function used below.

```

clean_func <- function(x, y){

  x1 <- x %>% select(subgroup, contains(y))

  # split up into mean and se
  my_se <- x1 %>% select(subgroup, starts_with("se."))
  my_mean <- x1 %>% select(subgroup, !starts_with("se."))

  # transform both datasets to long
  my_se_long <- my_se %>% pivot_longer(cols = starts_with("se."),
                                     names_to = "food",
                                     names_prefix = c("se."),
                                     values_to = paste0(y, "_se"))

  my_mean_long <- my_mean %>% pivot_longer(cols = !subgroup,
                                           names_to = "food",
                                           values_to = paste0(y, "_mean"))

  # join
  allfoods_bysub_long <- left_join(my_mean_long, my_se_long, by = c("subgroup", "f

  # fix names
  dat <- allfoods_bysub_long %>%
    mutate(food_type = case_when(grepl("fah", food) ~ "Grocery",
                                  grepl("fafh", food) ~ "Non-Grocery",
                                  grepl("total", food) ~ "Total")) %>%
    mutate(food = gsub(".*_sum_", "", food)) %>%
    arrange(subgroup, food)

  print(dat)

}

```

Import the data that were cleaned in the last few chapters.

```
# diet intake data
# both_days <- read_rds("data_inputs/IMPACT_FACTORS/temp_data/both_days15_env.rds"

# price data
price_dat <- read_rds("data_inputs/IMPACT_FACTORS/temp_data/price_input_dat.rds")

# environment and social data
enviro_dat <- read_rds("data_inputs/IMPACT_FACTORS/temp_data/enviro_input_dat.rds"
```

6.2 Cost Impact Factors

6.2.1 Cost Impact Factors, by Food, by Subgroup

Join the price data with the NHANES subgroup data.

```
# full join
price_dat1 <- full_join(price_dat, subgroup_dat, by = c("seqn" = "SEQN"))
```

Define the “survey design” for the cost dataset. Survey designs are a unique part of the R ‘survey’ package. Set the appropriate weight variables.

Then, subset the dataset to only include participants who are eligible to be in the sample (i.e., `inAnalysis == 1`). **Chapter XXX** explains more about these weight and `inAnalysis` variables.

```

# Define survey design for cost dataset
my_cost_svy <- svydesign(data=price_dat1,
                        id=~SDMVPSU, # Masked Variance Unit Pseudo-PSU
                        strata=~SDMVSTRA, # Masked Variance Unit Pseudo-Stratum
                        weights=~wtnew, # New sample weight
                        nest=TRUE)

# Create a survey design object for the subset of interest
my_cost_svy_sub <- subset(my_cost_svy, inAnalysis==1)

```

6.2.1.1 Calculate Daily Mean Cost, For The Whole Sample

Apply the 'svymean' function in order to calculate the mean daily cost for each food group, using the entire sample.

```

allfoods_cost <- svymean(reformulate(names(price_dat1) %>% str_subset("price")),
                        my_cost_svy_sub) %>%

as.data.frame() %>%

rownames_to_column(var = "var") %>%

pivot_wider(names_from = var,
            values_from = c(mean, SE)) %>%

mutate(subgroup = 0) %>%

relocate(subgroup)

# fix colnames
colnames(allfoods_cost) <- gsub("mean_", "", colnames(allfoods_cost))
colnames(allfoods_cost) <- gsub("SE_", "se.", colnames(allfoods_cost))

```

6.2.1.2 Calculate Daily Mean Consumed, Inedible, and Wasted Amounts (FNDDS-Level), For The Whole Sample

Apply the 'svymean' function in order to calculate the mean daily consumed, inedible, and wasted amounts of each food group, using the entire sample.

6.2.1.2.1 Consumed

```
allfoods_cost_consumed <- svymean(reformulate(names(price_dat1) %>% str_subset("cc
                                my_cost_svy_sub) %>%

as.data.frame() %>%
rownames_to_column(var = "var") %>%
pivot_wider(names_from = var,
             values_from = c(mean, SE)) %>%
mutate(subgroup = 0) %>%
relocate(subgroup)

# fix colnames
colnames(allfoods_cost_consumed) <- gsub("mean_", "", colnames(allfoods_cost_consu
colnames(allfoods_cost_consumed) <- gsub("SE_", "se.", colnames(allfoods_cost_cons
```

6.2.1.2.2 Inedible

```
allfoods_cost_inedible <- svymean(reformulate(names(price_dat1) %>% str_subset("ir
                                my_cost_svy_sub) %>%

as.data.frame() %>%
rownames_to_column(var = "var") %>%
pivot_wider(names_from = var,
             values_from = c(mean, SE)) %>%
mutate(subgroup = 0) %>%
relocate(subgroup)

# fix colnames
colnames(allfoods_cost_inedible) <- gsub("mean_", "", colnames(allfoods_cost_inedi
colnames(allfoods_cost_inedible) <- gsub("SE_", "se.", colnames(allfoods_cost_inec
```

6.2.1.2.3 Wasted

```
allfoods_cost_wasted <- svymean(reformulate(names(price_dat1) %>% str_subset("wast
                                my_cost_svy_sub) %>%

                                as.data.frame() %>%
                                rownames_to_column(var = "var") %>%
                                pivot_wider(names_from = var,
                                              values_from = c(mean, SE)) %>%
                                mutate(subgroup = 0) %>%
                                relocate(subgroup)

# fix colnames
colnames(allfoods_cost_wasted) <- gsub("mean_", "", colnames(allfoods_cost_wasted))
colnames(allfoods_cost_wasted) <- gsub("SE_", "se.", colnames(allfoods_cost_wasted))
```

6.2.1.3 Calculate Daily Mean Cost, For All 48 Subgroups

Apply the 'svyby' function in order to calculate the mean daily cost for each food group, separately for each of the 48 population subgroups.

```
allfoods_cost_bysub <- svyby(reformulate(names(price_dat1) %>% str_subset("price")
                                ~subgroup,
                                my_cost_svy_sub,
                                svymean)
```

Combine the mean cost values for the whole sample (i.e., subgroup 0) with the mean cost values for subgroups 1-48.

```
# bind with subgroup 0
allfoods_cost_bysub1 <- rbind(allfoods_cost, allfoods_cost_bysub)
```

6.2.1.4 Calculate Daily Mean Consumed, Inedible, and Wasted Amounts

(FNDDS-Level), For All 48 Subgroups

Apply the 'svyby' function in order to calculate the mean daily consumed, inedible, and wasted amounts of each food group, separately for each of the 48 population subgroups.

Then, combine the **mean values** for the whole sample (i.e., subgroup 0) with the mean values for subgroups 1-48.

6.2.1.4.1 Consumed

```
allfoods_cost_consumed_bysub <- svyby(reformulate(names(price_dat1) %>% str_subset
  ~subgroup,
  my_cost_svy_sub,
  svymean)
```

bind with subgroup 0

```
allfoods_cost_consumed_bysub1 <- rbind(allfoods_cost_consumed, allfoods_cost_consu
```

6.2.1.4.2 Inedible

```
allfoods_cost_inedible_bysub <- svyby(reformulate(names(price_dat1) %>% str_subset
  ~subgroup,
  my_cost_svy_sub,
  svymean)
```

bind with subgroup 0

```
allfoods_cost_inedible_bysub1 <- rbind(allfoods_cost_inedible, allfoods_cost_inedi
```


6.2.1.4.3 Wasted

```
allfoods_cost_wasted_bysub <- svyby(reformulate(names(price_dat1) %>% str_subset("
                                ~subgroup,
                                my_cost_svy_sub,
                                svymean)

# bind with subgroup 0
allfoods_cost_wasted_bysub1 <- rbind(allfoods_cost_wasted, allfoods_cost_wasted_by
```

6.2.1.5 Clean Output Data

Apply the cleaning function (defined above) to the mean cost dataset.

```
# APPLY FUNCTION TO COST DAT
cost_dat <- clean_func(x = allfoods_cost_bysub1, y = "price")
```

```
## # A tibble: 5,292 × 5
```

##	subgroup	food	price_mean	price_se	food_type
##	<dbl>	<chr>	<dbl>	<dbl>	<chr>
## 1	0	Consumed_added_sugar_0	0.164	0.00690	Total
## 2	0	Consumed_added_sugar_0	0.0826	0.00345	Grocery
## 3	0	Consumed_added_sugar_0	0.0811	0.00494	Non-Grocery
## 4	0	Consumed_babyfood_0	0.0000216	0.0000184	Total
## 5	0	Consumed_babyfood_0	0.0000216	0.0000184	Grocery
## 6	0	Consumed_babyfood_0	0	0	Non-Grocery
## 7	0	Consumed_babyfood_1	0.00117	0.00116	Total
## 8	0	Consumed_babyfood_1	0.00117	0.00116	Grocery
## 9	0	Consumed_babyfood_1	0	0	Non-Grocery
## 10	0	Consumed_dairy_tot_0	0.755	0.0244	Total

```
## # i 5,282 more rows
```

MORE MANUAL CLEANING

```
cost_dat1 <-
  cost_dat %>% mutate(intake_type = case_when(grepl("Consumed", food) ~ "Consumed"
                                              grepl("Wasted", food) ~ "Wasted",
                                              grepl("Inedible", food) ~ "Inedible"))
  mutate(food = gsub("Consumed_|Wasted_|Inedible_", "", food))

# MORE MANUAL CLEANING

cost_dat2 <-
  cost_dat1 %>% mutate(mixed_dish = case_when(grepl("_0", food) ~ "Non-Mixed",
                                              grepl("_1", food) ~ "Mixed")) %>%
  mutate(food = gsub("_0|_1", "", food)) %>%
  relocate(intake_type, food_type, mixed_dish, .after = food) %>%
  arrange(intake_type, subgroup, food, food_type, mixed_dish)

# examine

cost_dat2 %>% filter(food_type == "Total") %>% head()
```

A tibble: 6 × 7

##	subgroup	food	intake_type	food_type	mixed_dish	price_mean	price_se
##	<dbl>	<chr>	<chr>	<chr>	<chr>	<dbl>	<dbl>
## 1	0	added_sugar	Consumed	Total	Non-Mixed	0.164	0.00690
## 2	0	babyfood	Consumed	Total	Mixed	0.00117	0.00116
## 3	0	babyfood	Consumed	Total	Non-Mixed	0.0000216	0.0000184
## 4	0	dairy_tot	Consumed	Total	Non-Mixed	0.755	0.0244
## 5	0	fruit_exc_juice	Consumed	Total	Mixed	0.0180	0.00207
## 6	0	fruit_exc_juice	Consumed	Total	Non-Mixed	0.420	0.0170

```
# pivot to wide
cost_dat3 <-
  cost_dat2 %>% pivot_wider(names_from = intake_type,
                           values_from = c(price_mean, price_se)) %>%
  arrange(subgroup, food, food_type)
```

Apply the cleaning function (defined above) to the mean consumed, inedible, and wasted datasets.

```
# APPLY FUNCTION TO INTAKE DATASETS
cost_consumed_dat <- clean_func(x = allfoods_cost_consumed_bysub1, y = "consumed")
```

```
## # A tibble: 5,292 × 5
##   subgroup food          consumed_mean consumed_se food_type
##   <dbl> <chr>          <dbl>         <dbl> <chr>
## 1      0 added_sugar_0      16.8          0.571 Total
## 2      0 added_sugar_0      11.7          0.454 Grocery
## 3      0 added_sugar_0       5.11          0.304 Non-Grocery
## 4      0 babyfood_0        0.00712        0.00653 Total
## 5      0 babyfood_0        0.00712        0.00653 Grocery
## 6      0 babyfood_0         0             0 Non-Grocery
## 7      0 babyfood_1        0.185          0.184 Total
## 8      0 babyfood_1        0.185          0.184 Grocery
## 9      0 babyfood_1         0             0 Non-Grocery
## 10     0 dairy_tot_0       186.           5.90 Total
## # i 5,282 more rows
```

```
cost_wasted_dat <- clean_func(x = allfoods_cost_wasted_bysub1, y = "wasted")
```

```
## # A tibble: 5,292 × 5
##   subgroup food          wasted_mean wasted_se food_type
##   <dbl> <chr>          <dbl>      <dbl> <chr>
## 1      0 added_sugar_0    7.16      0.294 Total
## 2      0 added_sugar_0    5.04      0.254 Grocery
## 3      0 added_sugar_0    2.13      0.124 Non-Grocery
## 4      0 babyfood_0      0.000775 0.000724 Total
## 5      0 babyfood_0      0.000775 0.000724 Grocery
## 6      0 babyfood_0      0          0      Non-Grocery
## 7      0 babyfood_1      0.0449    0.0447 Total
## 8      0 babyfood_1      0.0449    0.0447 Grocery
## 9      0 babyfood_1      0          0      Non-Grocery
## 10     0 dairy_tot_0      54.1      1.73   Total
## # i 5,282 more rows
```

```
cost_inedible_dat <- clean_func(x = allfoods_cost_inedible_bysub1, y = "inedible")
```

```
## # A tibble: 5,292 × 5
##   subgroup food          inedible_mean inedible_se food_type
##   <dbl> <chr>          <dbl>      <dbl> <chr>
## 1      0 added_sugar_0    0.217      0.0228 Total
## 2      0 added_sugar_0    0.172      0.0193 Grocery
## 3      0 added_sugar_0    0.0450     0.00874 Non-Grocery
## 4      0 babyfood_0      0          0      Total
## 5      0 babyfood_0      0          0      Grocery
## 6      0 babyfood_0      0          0      Non-Grocery
## 7      0 babyfood_1      0.0475     0.0473 Total
## 8      0 babyfood_1      0.0475     0.0473 Grocery
## 9      0 babyfood_1      0          0      Non-Grocery
## 10     0 dairy_tot_0      1.72      0.251   Total
## # i 5,282 more rows
```

```

# join all 3
cost_intake_dat <- left_join(cost_consumed_dat, cost_wasted_dat, by = c("subgroup"
  left_join(cost_inedible_dat, by = c("subgroup", "food", "food_type"))

# create mixed dish variable
cost_intake_dat1 <- cost_intake_dat %>%
  mutate(mixed_dish = case_when(grepl("_0", food) ~ "Non-Mixed",
                                grepl("_1", food) ~ "Mixed")) %>%
  mutate(food = gsub("_0|_1", "", food))

# join
cost_final_dat <-
  left_join(cost_dat3, cost_intake_dat1, by = c("subgroup", "food", "food_type", "
  rename(fndds_consumed_mean = consumed_mean,
         fndds_consumed_se = consumed_se,
         fndds_wasted_mean = wasted_mean,
         fndds_wasted_se = wasted_se,
         fndds_inedible_mean = inedible_mean,
         fndds_inedible_se = inedible_se)

```

6.2.2 Cost Impact Factors, Per 100 Grams and Per 1 DGA Serving

First, import the conversion units.

```

units <- read_csv(paste("data_inputs/FINAL/cleaned_raw_data/unit_conversions_", my
  select(Food_group, Conversion_to_grams)

```

```
## Rows: 41 Columns: 4
## — Column specification —————
## Delimiter: ","
## chr (3): Food_group, DGA_unit, Equation
## dbl (1): Conversion_to_grams
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
# merge with units
cost_final_dat1 <- cost_final_dat %>%
  left_join(units, by = c("food" = "Food_group"))

cost_final_dat1 %>% filter(is.na(Conversion_to_grams)) %>% head()
```

```
## # A tibble: 0 × 13
## # i 13 variables: subgroup <dbl>, food <chr>, food_type <chr>, mixed_dish <chr>, price_m
## #   price_se_Consumed <dbl>, fndds_consumed_mean <dbl>, fndds_consumed_se <dbl>, fndds_v
## #   fndds_wasted_se <dbl>, fndds_inedible_mean <dbl>, fndds_inedible_se <dbl>, Conversic
```

Then, calculate the cost impact factors (i) per 100 grams and (ii) per DGA unit.

To calculate the cost per 100 grams, you divide the daily mean cost of the food group by the daily amount of food consumed, and then multiply by 100.

```
# calculate impact factors
cost_final_dat2 <- cost_final_dat1 %>%
  rowwise() %>%
  mutate(costper100gram_consumed = ifelse(fndds_consumed_mean == 0, 0, (price_mean
    costperDGA_consumed = costper100gram_consumed * (Conversion_to_grams / 1000)

# check
cost_final_dat2 %>% select(subgroup, food, food_type, mixed_dish, costperDGA_consumed)
```

```
## # A tibble: 6 × 5
## # Rowwise:
##   subgroup food      food_type  mixed_dish costperDGA_consumed
##   <dbl> <chr>      <chr>      <chr>      <dbl>
## 1      0 added_sugar Grocery    Non-Mixed    0.0297
## 2      0 added_sugar Non-Grocery Non-Mixed    0.0667
## 3      0 added_sugar Total      Non-Mixed    0.0410
## 4      0 babyfood  Grocery    Mixed        0
## 5      0 babyfood  Grocery    Non-Mixed    0
## 6      0 babyfood  Non-Grocery Mixed        0
```

6.3 Environmental & Social Impact Factors

6.3.1 Environmental & Social Impact Factors, by Food, by Subgroup

Essentially the same steps that were used to calculate the cost impact factors are also used to calculate the environmental and social impact factors.

First, define the “survey design” for the enviro dataset. Survey designs are a unique part of the R ‘survey’ package. Set the appropriate weight variables.

Then, subset the dataset to only include participants who are eligible to be in the sample (i.e., `inAnalysis == 1`). **Chapter XXX** explains more about these weight and `inAnalysis` variables.

```
# join
enviro_dat1 <- full_join(enviro_dat, subgroup_dat, by = c("seqn" = "SEQN"))

# Define survey design for ghg dataset
my_enviro_svy <- svydesign(data=enviro_dat1,
                        id=~SDMVPSU, # Masked Variance Unit Pseudo-PSU
                        strata=~SDMVSTRA, # Masked Variance Unit Pseudo-Stratum
                        weights=~wtnew, # New sample weight
                        nest=TRUE)

# Create a survey design object for the subset of interest
my_enviro_svy_sub <- subset(my_enviro_svy, inAnalysis==1)
```



6.3.1.1 Calculate Daily Mean Environment and Social Impacts, (i) For

The Whole Sample, and (ii) For All 48 Subgroups

6.3.1.1.1 Greenhouse Gas Emissions (GHG)

```
# CALCULATE GHG IMPACT (WHOLE SAMPLE)

allfoods_ghg <- svymean(reformulate(names(enviro_dat1) %>% str_subset("ghg")),
                        my_enviro_svy_sub) %>%

  as.data.frame() %>%
  rownames_to_column(var = "var") %>%
  pivot_wider(names_from = var,
              values_from = c(mean, SE)) %>%
  mutate(subgroup = 0) %>%
  relocate(subgroup)

# fix colnames
colnames(allfoods_ghg) <- gsub("mean_", "", colnames(allfoods_ghg))
colnames(allfoods_ghg) <- gsub("SE_", "se.", colnames(allfoods_ghg))

# CALCULATE GHG IMPACT, BY SUBGROUP

allfoods_ghg_bysub <- svyby(reformulate(names(enviro_dat1) %>% str_subset("ghg")),
                           ~subgroup,
                           my_enviro_svy_sub,
                           svymean)

# bind with subgroup 0
allfoods_ghg_bysub1 <- rbind(allfoods_ghg, allfoods_ghg_bysub)
```

6.3.1.1.2 Cumulative Energy Demand (CED)

```
# CALCULATE CED IMPACT (WHOLE SAMPLE)

allfoods_ced <- svymean(reformulate(names(enviro_dat1) %>% str_subset("ced")),
                        my_enviro_svy_sub) %>%

  as.data.frame() %>%

  rownames_to_column(var = "var") %>%

  pivot_wider(names_from = var,
              values_from = c(mean, SE)) %>%

  mutate(subgroup = 0) %>%

  relocate(subgroup)


# fix colnames

colnames(allfoods_ced) <- gsub("mean_", "", colnames(allfoods_ced))
colnames(allfoods_ced) <- gsub("SE_", "se.", colnames(allfoods_ced))


# CALCULATE CED IMPACT, BY SUBGROUP

allfoods_ced_bysub <- svyby(reformulate(names(enviro_dat1) %>% str_subset("ced")),
                           ~subgroup,
                           my_enviro_svy_sub,
                           svymean)


# bind with subgroup 0

allfoods_ced_bysub1 <- rbind(allfoods_ced, allfoods_ced_bysub)
```

6.3.1.1.3 Water Scarcity

```
# CALCULATE WATER IMPACT (WHOLE SAMPLE)

allfoods_water <- svymean(reformulate(names(enviro_dat1) %>% str_subset("^water"))
                          my_enviro_svy_sub) %>%

  as.data.frame() %>%

  rownames_to_column(var = "var") %>%

  pivot_wider(names_from = var,
              values_from = c(mean, SE)) %>%

  mutate(subgroup = 0) %>%

  relocate(subgroup)


# fix colnames

colnames(allfoods_water) <- gsub("mean_", "", colnames(allfoods_water))
colnames(allfoods_water) <- gsub("SE_", "se.", colnames(allfoods_water))


# CALCULATE WATER IMPACT, BY SUBGROUP

allfoods_water_bysub <- svyby(reformulate(names(enviro_dat1) %>% str_subset("^water
~subgroup,
my_enviro_svy_sub,
svymean)


# bind with subgroup 0

allfoods_water_bysub1 <- rbind(allfoods_water, allfoods_water_bysub)
```

6.3.1.1.4 Bluewater Use

```
# CALCULATE BLUEWATER IMPACT (WHOLE SAMPLE)

allfoods_bluewater <- svymean(reformulate(names(enviro_dat1) %>% str_subset("bluew
                                my_enviro_svy_sub) %>%

                                as.data.frame() %>%

                                rownames_to_column(var = "var") %>%

                                pivot_wider(names_from = var,

                                              values_from = c(mean, SE)) %>%

                                mutate(subgroup = 0) %>%

                                relocate(subgroup)

# fix colnames

colnames(allfoods_bluewater) <- gsub("mean_", "", colnames(allfoods_bluewater))
colnames(allfoods_bluewater) <- gsub("SE_", "se.", colnames(allfoods_bluewater))

# CALCULATE BLUEWATER IMPACT, BY SUBGROUP

allfoods_bluewater_bysub <- svyby(reformulate(names(enviro_dat1) %>% str_subset("t
                                ~subgroup,

                                my_enviro_svy_sub,

                                svymean)

# bind with subgroup 0

allfoods_bluewater_bysub1 <- rbind(allfoods_bluewater, allfoods_bluewater_bysub)
```

6.3.1.1.5 Forced Labor Risk (FL)

```
# CALCULATE FL IMPACT (WHOLE SAMPLE)

allfoods_fl <- svymean(reformulate(names(enviro_dat1) %>% str_subset("fl")),
                      my_enviro_svy_sub) %>%

  as.data.frame() %>%

  rownames_to_column(var = "var") %>%

  pivot_wider(names_from = var,
              values_from = c(mean, SE)) %>%

  mutate(subgroup = 0) %>%

  relocate(subgroup)

# fix colnames

colnames(allfoods_fl) <- gsub("mean_", "", colnames(allfoods_fl))
colnames(allfoods_fl) <- gsub("SE_", "se.", colnames(allfoods_fl))

# CALCULATE FL IMPACT, BY SUBGROUP

allfoods_fl_bysub <- svyby(reformulate(names(enviro_dat1) %>% str_subset("fl")),
                          ~subgroup,
                          my_enviro_svy_sub,
                          svymean)

# bind with subgroup 0

allfoods_fl_bysub1 <- rbind(allfoods_fl, allfoods_fl_bysub)
```

6.3.1.2 Calculate Daily Mean Consumed, Inedible, and Wasted Amounts (FCID-Level), For The Whole Sample

Apply the 'svymean' function in order to calculate the mean daily consumed, inedible, and wasted amounts of each food group, using the entire sample.

6.3.1.2.1 Consumed

```
# consumed

allfoods_enviro_consumed <- svymean(reformulate(names(enviro_dat1) %>% str_subset(
  my_enviro_svy_sub) %>%

  as.data.frame() %>%
  rownames_to_column(var = "var") %>%
  pivot_wider(names_from = var,
    values_from = c(mean, SE)) %>%
  mutate(subgroup = 0) %>%
  relocate(subgroup)

# fix colnames

colnames(allfoods_enviro_consumed) <- gsub("mean_", "", colnames(allfoods_enviro_c
colnames(allfoods_enviro_consumed) <- gsub("SE_", "se.", colnames(allfoods_enviro_
```

6.3.1.2.2 Inedible

```
# inedible

allfoods_enviro_inedible <- svymean(reformulate(names(enviro_dat1) %>% str_subset(
  my_enviro_svy_sub) %>%

  as.data.frame() %>%
  rownames_to_column(var = "var") %>%
  pivot_wider(names_from = var,
    values_from = c(mean, SE)) %>%
  mutate(subgroup = 0) %>%
  relocate(subgroup)

# fix colnames

colnames(allfoods_enviro_inedible) <- gsub("mean_", "", colnames(allfoods_enviro_i
colnames(allfoods_enviro_inedible) <- gsub("SE_", "se.", colnames(allfoods_enviro_
```

6.3.1.2.3 Wasted

```
# wasted

allfoods_enviro_wasted <- svymean(reformulate(names(enviro_dat1) %>% str_subset("w
                                my_enviro_svy_sub) %>%

as.data.frame() %>%
rownames_to_column(var = "var") %>%
pivot_wider(names_from = var,
             values_from = c(mean, SE)) %>%
mutate(subgroup = 0) %>%
relocate(subgroup)

# fix colnames

colnames(allfoods_enviro_wasted) <- gsub("mean_", "", colnames(allfoods_enviro_was
colnames(allfoods_enviro_wasted) <- gsub("SE_", "se.", colnames(allfoods_enviro_wa
```

6.3.1.3 Calculate Daily Mean Consumed, Inedible, and Wasted Amounts (FCID-Level), For All 48 Subgroups

Apply the 'svyby' function in order to calculate the mean daily consumed, inedible, and wasted amounts of each food group, separately for each of the 48 population subgroups.

Then, combine the mean values for the whole sample (i.e., subgroup 0) with the mean values for subgroups 1-48.

6.3.1.3.1 Consumed

```
allfoods_enviro_consumed_bysub <- svyby(reformulate(names(enviro_dat1) %>% str_sub
                                ~subgroup,
                                my_enviro_svy_sub,
                                svymean)

# bind with subgroup 0

allfoods_enviro_consumed_bysub1 <- rbind(allfoods_enviro_consumed, allfoods_enviro
```

6.3.1.3.2 Inedible

```
# inedible

allfoods_enviro_inedible_bysub <- svyby(reformulate(names(enviro_dat1) %>% str_sub(
  ~subgroup,
  my_enviro_svy_sub,
  svymean)

# bind with subgroup 0

allfoods_enviro_inedible_bysub1 <- rbind(allfoods_enviro_inedible, allfoods_enviro_inedible_bysub1)
```

6.3.1.3.3 Wasted

```
# wasted

allfoods_enviro_wasted_bysub <- svyby(reformulate(names(enviro_dat1) %>% str_sub(
  ~subgroup,
  my_enviro_svy_sub,
  svymean)

# bind with subgroup 0

allfoods_enviro_wasted_bysub1 <- rbind(allfoods_enviro_wasted, allfoods_enviro_wasted_bysub1)
```

6.3.1.4 Clean Output Data

Apply the cleaning function (defined above) to the mean enviro datasets.

```
# APPLY TO ENVIRO DATA

ghg_dat <- clean_func(x = allfoods_ghg_bysub1, y = "ghg") %>% select(-food_type)
```



```
## # A tibble: 1,176 × 5
##   subgroup food                ghg_mean ghg_se food_type
##   <dbl> <chr>                <dbl>   <dbl> <chr>
## 1      0 Consumed_added_sugar  0.0504 0.000994 <NA>
## 2      0 Consumed_babyfood      0      0      <NA>
## 3      0 Consumed_coffee_tea    0.0453 0.00142  <NA>
## 4      0 Consumed_dairy_cow     0.326  0.00671  <NA>
## 5      0 Consumed_dairy_soy     0.00251 0.000218 <NA>
## 6      0 Consumed_fruit_exc_juice 0.0617 0.00371  <NA>
## 7      0 Consumed_fruit_juice    0.0536 0.00199  <NA>
## 8      0 Consumed_gr_refined     0.0620 0.00107  <NA>
## 9      0 Consumed_gr_whole       0.0169 0.000426 <NA>
## 10     0 Consumed_leg_tot        0.0158 0.000514 <NA>
## # i 1,166 more rows
```

```
ced_dat <- clean_func(x = allfoods_ced_bysub1, y = "ced") %>% select(-food_type)
```

```
## # A tibble: 1,176 × 5
##   subgroup food                ced_mean ced_se food_type
##   <dbl> <chr>                <dbl>   <dbl> <chr>
## 1      0 Consumed_added_sugar  0.492  0.0111  <NA>
## 2      0 Consumed_babyfood      0      0      <NA>
## 3      0 Consumed_coffee_tea    0.470  0.0146  <NA>
## 4      0 Consumed_dairy_cow     1.76   0.0362  <NA>
## 5      0 Consumed_dairy_soy     0.0164 0.00142 <NA>
## 6      0 Consumed_fruit_exc_juice 0.468  0.0204  <NA>
## 7      0 Consumed_fruit_juice    0.422  0.0143  <NA>
## 8      0 Consumed_gr_refined     0.354  0.00559 <NA>
## 9      0 Consumed_gr_whole       0.107  0.00230 <NA>
## 10     0 Consumed_leg_tot        0.0956 0.00270 <NA>
## # i 1,166 more rows
```

```
water_dat <- clean_func(x = allfoods_water_bysub1, y = "water") %>% select(-food_t
```

```
## # A tibble: 1,176 × 5
```

```
##   subgroup food                water_mean water_se food_type
##   <dbl> <chr>                <dbl>    <dbl> <chr>
## 1      0 Consumed_added_sugar    35.8    0.719 <NA>
## 2      0 Consumed_babyfood        0        0 <NA>
## 3      0 Consumed_coffee_tea    51.5    1.62 <NA>
## 4      0 Consumed_dairy_cow    91.4    1.88 <NA>
## 5      0 Consumed_dairy_soy     0.219   0.0190 <NA>
## 6      0 Consumed_fruit_exc_juice 217.    11.2 <NA>
## 7      0 Consumed_fruit_juice   140.     3.93 <NA>
## 8      0 Consumed_gr_refined    96.5    1.48 <NA>
## 9      0 Consumed_gr_whole     32.3    1.09 <NA>
## 10     0 Consumed_leg_tot      24.6    1.07 <NA>
## # i 1,166 more rows
```

```
bluewater_dat <- clean_func(x = allfoods_bluewater_bysub1, y = "bluewater") %>% se
```

```
## # A tibble: 1,176 × 5
```

	subgroup	food	bluewater_mean	bluewater_se	food_type
	<dbl>	<chr>	<dbl>	<dbl>	<chr>
## 1	0	Consumed_added_sugar	9.68	0.193	<NA>
## 2	0	Consumed_babyfood	0	0	<NA>
## 3	0	Consumed_coffee_tea	9.15	0.293	<NA>
## 4	0	Consumed_dairy_cow	21.8	0.449	<NA>
## 5	0	Consumed_dairy_soy	0.126	0.0109	<NA>
## 6	0	Consumed_fruit_exc_juice	29.4	1.49	<NA>
## 7	0	Consumed_fruit_juice	16.9	0.500	<NA>
## 8	0	Consumed_gr_refined	30.7	0.451	<NA>
## 9	0	Consumed_gr_whole	6.66	0.171	<NA>
## 10	0	Consumed_leg_tot	5.87	0.213	<NA>

```
## # i 1,166 more rows
```

```
fl_dat <- clean_func(x = allfoods_fl_bysub1, y = "fl") %>% select(-food_type)
```

```
## # A tibble: 1,176 × 5
```

	subgroup	food	fl_mean	fl_se	food_type
	<dbl>	<chr>	<dbl>	<dbl>	<chr>
## 1	0	Consumed_added_sugar	0.0223	0.000390	<NA>
## 2	0	Consumed_babyfood	0	0	<NA>
## 3	0	Consumed_coffee_tea	0	0	<NA>
## 4	0	Consumed_dairy_cow	0.0412	0.000847	<NA>
## 5	0	Consumed_dairy_soy	0.00274	0.000238	<NA>
## 6	0	Consumed_fruit_exc_juice	0.0304	0.00147	<NA>
## 7	0	Consumed_fruit_juice	0.0264	0.000803	<NA>
## 8	0	Consumed_gr_refined	0.00801	0.000203	<NA>
## 9	0	Consumed_gr_whole	0.00205	0.0000627	<NA>
## 10	0	Consumed_leg_tot	0.00406	0.000141	<NA>

```
## # i 1,166 more rows
```

```

# join cleaned datasets
enviro_dat <-
  left_join(ghg_dat, ced_dat, by = c("subgroup", "food")) %>%
  left_join(water_dat, by = c("subgroup", "food")) %>%
  left_join(bluewater_dat, by = c("subgroup", "food")) %>%
  left_join(fl_dat, by = c("subgroup", "food"))

# more cleaning
enviro_dat1 <-
  enviro_dat %>% mutate(intake_type = case_when(grepl("Consumed", food) ~ "Consumed",
                                                grepl("Wasted", food) ~ "Wasted",
                                                grepl("Inedible", food) ~ "Inedible",
                                                TRUE ~ "Other"))
  mutate(food = gsub("Consumed_|Wasted_|Inedible_", "", food))

# pivot to wide
enviro_dat2 <-
  enviro_dat1 %>% pivot_wider(names_from = intake_type,
                             values_from = !c(subgroup, food, intake_type)) %>%
  arrange(subgroup, food)

```

Apply the cleaning function (defined above) to the mean consumed, inedible, and wasted datasets.

```

# APPLY FUNCTION TO INTAKE DAT
enviro_consumed_dat <- clean_func(x = allfoods_enviro_consumed_bysub1, y = "consumed")

```

```
## # A tibble: 1,176 × 5
##   subgroup food          consumed_mean consumed_se food_type
##   <dbl> <chr>          <dbl>      <dbl> <chr>
## 1      0 added_sugar      81.1      1.71 <NA>
## 2      0 babyfood        1.09      0.176 <NA>
## 3      0 coffee_tea       6.84      0.223 <NA>
## 4      0 dairy_cow      189.      5.04 <NA>
## 5      0 dairy_soy       9.72      0.843 <NA>
## 6      0 fruit_exc_juice 107.      4.03 <NA>
## 7      0 fruit_juice     57.0      1.80 <NA>
## 8      0 gr_refined     108.      1.31 <NA>
## 9      0 gr_whole       33.6      0.831 <NA>
## 10     0 leg_tot       21.5      0.577 <NA>
## # i 1,166 more rows
```

```
enviro_wasted_dat <- clean_func(x = allfoods_enviro_wasted_bysub1, y = "wasted") %
```

```
## # A tibble: 1,176 × 5
##   subgroup food          wasted_mean wasted_se food_type
##   <dbl> <chr>          <dbl>      <dbl> <chr>
## 1      0 added_sugar      41.3      0.887 <NA>
## 2      0 babyfood        0.153      0.0461 <NA>
## 3      0 coffee_tea         0          0 <NA>
## 4      0 dairy_cow      60.4      1.51 <NA>
## 5      0 dairy_soy       2.64      0.229 <NA>
## 6      0 fruit_exc_juice  84.4      5.04 <NA>
## 7      0 fruit_juice     8.06      0.326 <NA>
## 8      0 gr_refined     31.3      0.418 <NA>
## 9      0 gr_whole       8.20      0.189 <NA>
## 10     0 leg_tot       4.91      0.149 <NA>
## # i 1,166 more rows
```

```
enviro_inedible_dat <- clean_func(x = allfoods_enviro_inedible_bysub1, y = "inedit
```

```
## # A tibble: 1,176 × 5
```

```
##   subgroup food          inedible_mean inedible_se food_type
##   <dbl> <chr>          <dbl>         <dbl> <chr>
## 1      0 added_sugar      0             0      <NA>
## 2      0 babyfood        0.0475        0.0473 <NA>
## 3      0 coffee_tea      0             0      <NA>
## 4      0 dairy_cow       0             0      <NA>
## 5      0 dairy_soy       0             0      <NA>
## 6      0 fruit_exc_juice  87.3          6.06    <NA>
## 7      0 fruit_juice     0             0      <NA>
## 8      0 gr_refined      0             0      <NA>
## 9      0 gr_whole        5.30          0.245   <NA>
## 10     0 leg_tot          1.18          0.0663  <NA>
## # i 1,166 more rows
```

```

# join
enviro_intake_dat <- left_join(enviro_consumed_dat, enviro_wasted_dat, by = c("subgroup", "food"))
left_join(enviro_inedible_dat, by = c("subgroup", "food"))

# join enviro and intake dat
enviro_final_dat <-
  left_join(enviro_dat2, enviro_intake_dat, by = c("subgroup", "food")) %>%
  rename(fcid_consumed_mean = consumed_mean,
         fcid_consumed_se = consumed_se,
         fcid_wasted_mean = wasted_mean,
         fcid_wasted_se = wasted_se,
         fcid_inedible_mean = inedible_mean,
         fcid_inedible_se = inedible_se)

# Look at just inedible and wasted
enviro_final_dat %>%
  filter(subgroup == 0) %>%
  select(food, fcid_consumed_mean, fcid_wasted_mean, fcid_inedible_mean) %>%
  head()

```

```

## # A tibble: 6 × 4
##   food          fcid_consumed_mean fcid_wasted_mean fcid_inedible_mean
##   <chr>          <dbl>          <dbl>          <dbl>
## 1 added_sugar    81.1            41.3            0
## 2 babyfood       1.09            0.153           0.0475
## 3 coffee_tea     6.84            0               0
## 4 dairy_cow     189.            60.4            0
## 5 dairy_soy      9.72            2.64            0
## 6 fruit_exc_juice 107.            84.4            87.3

```

```

# something funky happening with inedible amts of meat
# update: this is correct, fcid weight basis for meat is boneless meat so inedible

```

6.3.2 Environment/Social Impact Factors, Per 100 Grams and Per 1 DGA Serving

First, join with the conversion units.

```
# merge with units
enviro_final_dat1 <- enviro_final_dat %>%
  left_join(units, by = c("food" = "Food_group"))

enviro_final_dat1 %>% filter(is.na(Conversion_to_grams)) #good

## # A tibble: 0 × 19
## # i 19 variables: subgroup <dbl>, food <chr>, ghg_mean_Consumed <dbl>, ghg_se_Consumed <
## #   ced_se_Consumed <dbl>, water_mean_Consumed <dbl>, water_se_Consumed <dbl>, bluewater
## #   bluewater_se_Consumed <dbl>, fl_mean_Consumed <dbl>, fl_se_Consumed <dbl>, fcid_cons
## #   fcid_consumed_se <dbl>, fcid_wasted_mean <dbl>, fcid_wasted_se <dbl>, fcid_inedible_
## #   fcid_inedible_se <dbl>, Conversion_to_grams <dbl>
```

Then, calculate the environment/social impact factors (i) per 100 grams and (ii) per DGA unit.

To calculate the impacts (e.g., GHG) per 100 grams, you divide the daily mean GHG of the food group by the daily amount of food consumed AND inedible, and then multiply by 100.

Note: The variable “ghg_mean_Consumed” represents the GHG impact of consumed and inedible food (because the impact of both the consumed and inedible amounts is included in the original datafield impact factors). Therefore, the denominator of the equation is the sum of consumed food AND inedible food.


```

enviro_final_dat2 <- enviro_final_dat1 %>%
  rowwise() %>%
  mutate(#GHG
    GHGper100gram_consumed = ifelse(fcid_consumed_mean == 0, 0, (ghg_mean_Cor
    GHGperDGA_consumed = GHGper100gram_consumed * (Conversion_to_grams / 100)

  #CED
  CEDper100gram_consumed = ifelse(fcid_consumed_mean == 0, 0, (ced_mean_Cor
  CEDperDGA_consumed = CEDper100gram_consumed * (Conversion_to_grams / 100)

  #WATER
  WATERper100gram_consumed = ifelse(fcid_consumed_mean == 0, 0, (water_mear
  WATERperDGA_consumed = WATERper100gram_consumed * (Conversion_to_grams /

  #BLUEWATER
  BLUEWATERper100gram_consumed = ifelse(fcid_consumed_mean == 0, 0, (bluewa
  BLUEWATERperDGA_consumed = BLUEWATERper100gram_consumed * (Conversion_to_

  #FORCED LABOR
  FLper100gram_consumed = ifelse(fcid_consumed_mean == 0, 0, (fl_mean_Consum
  FLperDGA_consumed = FLper100gram_consumed * (Conversion_to_grams / 100))

  select(-Conversion_to_grams)

```

6.4 Finalize Impact Factors

6.4.1 Merge Cost, Environment, and Social Impact Factors

Merge all of the impact factors into one dataset.

```

# cost impact factors
cost_IFs <- cost_final_dat2 %>%
  rename(costper100g = costper100gram_consumed,
         costperDGA = costperDGA_consumed) %>%
  filter(!(food %in% c("babyfood", "coffee_tea", "other", "water")))

# environment impact factors
enviro_IFs <- enviro_final_dat2 %>%
  filter(!(food %in% c("babyfood", "coffee_tea", "other", "water"))) %>%
  rename(GHGperDGA = GHGperDGA_consumed,
         GHGper100g = GHGper100gram_consumed,

         CEDperDGA = CEDperDGA_consumed,
         CEDper100g = CEDper100gram_consumed,

         WATERperDGA = WATERperDGA_consumed,
         WATERper100g = WATERper100gram_consumed,

         BLUEWATERperDGA = BLUEWATERperDGA_consumed,
         BLUEWATERper100g = BLUEWATERper100gram_consumed,

         FLperDGA = FLperDGA_consumed,
         FLper100g = FLper100gram_consumed)

```

6.5 Calculate Inedible and Wasted Proportions

6.5.1 Environment/Social

The following proportions were calculated at the FCID-level:

- Wasted Proportion = Mean Wasted Amount / Mean Edible Amount
- Inedible Proportion = Mean Inedible Amount / Mean Purchased Amount

```

enviro_IFs1 <- enviro_IFs %>%
  rowwise() %>%
  mutate(# coefficients
    wasted_coef_fcid = fcid_wasted_mean / fcid_consumed_mean,
    inedible_coef_fcid = fcid_inedible_mean / fcid_consumed_mean,

    # amounts
    fcid_purchased_mean = sum(fcid_consumed_mean, fcid_wasted_mean, fcid_inedible_mean),
    fcid_edible_mean = sum(fcid_consumed_mean, fcid_wasted_mean),

    # proportions
    wasted_prop_fcid = fcid_wasted_mean / fcid_edible_mean,
    inedible_prop_fcid = fcid_inedible_mean / fcid_purchased_mean,
    consumed_prop_fcid = fcid_consumed_mean / fcid_edible_mean,
    prop_sum = sum(wasted_prop_fcid, consumed_prop_fcid)) ## change to sumir

# is sum = 1?
table(enviro_IFs1$prop_sum, useNA = "always") #good

```

##

1 <NA>

980 0

```
# create dataset with just IFs
```

```
enviro_IFs_sub <- enviro_IFs1 %>%
```

```
  select(subgroup, food,
         wasted_coef_fcid,
         wasted_prop_fcid,
         inedible_coef_fcid,
         inedible_prop_fcid,
         contains("perDGA"))
```

```
# Look at a few foods
```

```
enviro_IFs_sub %>% filter(food == "gr_whole") %>% head()
```

```
## # A tibble: 6 × 11
```

```
## # Rowwise:
```

```
##   subgroup food   wasted_coef_fcid wasted_prop_fcid inedible_coef_fcid inedible_prop_fcid
##   <dbl> <chr>         <dbl>           <dbl>           <dbl>           <dbl>
## 1      0 gr_w...      0.244           0.196           0.158           0.113
## 2      1 gr_w...      0.256           0.204           0.115           0.0841
## 3      2 gr_w...      0.256           0.204           0.106           0.0778
## 4      3 gr_w...      0.249           0.199           0.141           0.101
## 5      4 gr_w...      0.255           0.203           0.168           0.118
## 6      5 gr_w...      0.240           0.194           0.181           0.127
## # i 2 more variables: BLUEWATERperDGA <dbl>, FLperDGA <dbl>
```

```
enviro_IFs_sub %>% filter(food == "pf_redm_tot") %>% head()
```

```
## # A tibble: 6 × 11
## # Rowwise:
##   subgroup food   wasted_coef_fcic wasted_prop_fcic inedible_coef_fcic inedible_prop_fcic
##   <dbl> <chr>         <dbl>         <dbl>         <dbl>         <dbl>
## 1      0 pf_r...      0.306         0.234           0           0
## 2      1 pf_r...      0.310         0.237           0           0
## 3      2 pf_r...      0.314         0.239           0           0
## 4      3 pf_r...      0.295         0.228           0           0
## 5      4 pf_r...      0.305         0.234           0           0
## 6      5 pf_r...      0.306         0.234           0           0
## # i 2 more variables: BLUEWATERperDGA <dbl>, FLperDGA <dbl>
```

```
enviro_IFs_sub %>% filter(food == "veg_dg") %>% head()
```

```
## # A tibble: 6 × 11
## # Rowwise:
##   subgroup food   wasted_coef_fcic wasted_prop_fcic inedible_coef_fcic inedible_prop_fcic
##   <dbl> <chr>         <dbl>         <dbl>         <dbl>         <dbl>
## 1      0 veg_...      0.381         0.276         0.757         0.354
## 2      1 veg_...      0.288         0.223         0.568         0.306
## 3      2 veg_...      0.595         0.373         0.967         0.377
## 4      3 veg_...      0.317         0.241         0.828         0.386
## 5      4 veg_...      0.398         0.285         0.793         0.362
## 6      5 veg_...      0.304         0.233         0.722         0.356
## # i 2 more variables: BLUEWATERperDGA <dbl>, FLperDGA <dbl>
```

6.5.2 Cost

The following proportions were calculated at the FNDDS-level:

- Wasted Proportion = Mean Wasted Amount / Mean Edible Amount
- Inedible Proportion = Mean Inedible Amount / Mean Purchased Amount

```

cost_IFs1 <- cost_IFs %>%
  rowwise() %>%
  mutate(# coefficients
    wasted_coef_fndds = fndds_wasted_mean / fndds_consumed_mean,
    inedible_coef_fndds = fndds_inedible_mean / fndds_consumed_mean,

    # amounts
    fndds_purchased_mean = sum(fndds_consumed_mean, fndds_wasted_mean, fndds_
    fndds_edible_mean = sum(fndds_consumed_mean, fndds_wasted_mean),

    # proportions
    wasted_prop_fndds = ifelse(fndds_edible_mean == 0, 0, fndds_wasted_mean /
    inedible_prop_fndds = ifelse(fndds_purchased_mean == 0, 0, fndds_inedible
    consumed_prop_fndds = ifelse(fndds_edible_mean == 0, 0, fndds_consumed_me
    prop_sum_fndds = sum(wasted_prop_fndds, consumed_prop_fndds)) ##c hange

  table(cost_IFs1$prop_sum_fndds, useNA = "always")

```



```

##
##      0      1 <NA>
## 363 4194      0

```

```

# check
cost_IFs1 %>%
  filter(wasted_prop_fndds == 0) %>% head()

```

```
## # A tibble: 6 × 23
## # Rowwise:
##   subgroup food      food_type mixed_dish price_mean_Consumed price_se_Consumed fndds
##   <dbl> <chr>      <chr>      <chr>          <dbl>          <dbl>
## 1         1 fruit_juice Non-Groce... Mixed              0              0
## 2         1 veg_oth   Grocery     Mixed              0              0
## 3         1 veg_oth   Non-Groce... Mixed              0              0
## 4         1 veg_oth   Total       Mixed              0              0
## 5         2 fruit_juice Grocery     Mixed              0              0
## 6         2 fruit_juice Non-Groce... Mixed              0              0
## # i 15 more variables: fndds_wasted_mean <dbl>, fndds_wasted_se <dbl>, fndds_inedible_me
## #   fndds_inedible_se <dbl>, Conversion_to_grams <dbl>, costper100g <dbl>, costperDGA <c
## #   inedible_coef_fndds <dbl>, fndds_purchased_mean <dbl>, fndds_edible_mean <dbl>, wast
## #   inedible_prop_fndds <dbl>, consumed_prop_fndds <dbl>, prop_sum_fndds <dbl>
```

```
# subset
```

```
cost_IFs_sub <- cost_IFs1 %>%
  select(subgroup, food, food_type, mixed_dish,
         wasted_coef_fndds, wasted_prop_fndds,
         inedible_coef_fndds, inedible_prop_fndds,
         costperDGA)
```

```
# check a few
```

```
cost_IFs_sub %>% filter(food == "gr_whole") %>% head()
```

```
## # A tibble: 6 × 9
## # Rowwise:
##   subgroup food      food_type  mixed_dish wasted_coef_fndds wasted_prop_fndds inedible_
##   <dbl> <chr>    <chr>      <chr>          <dbl>          <dbl>
## 1      0 gr_whole Grocery    Mixed          0.287          0.223
## 2      0 gr_whole Grocery    Non-Mixed      0.174          0.148
## 3      0 gr_whole Non-Grocery Mixed          0.527          0.345
## 4      0 gr_whole Non-Grocery Non-Mixed      0.172          0.147
## 5      0 gr_whole Total      Mixed          0.325          0.245
## 6      0 gr_whole Total      Non-Mixed      0.174          0.148
## # i 1 more variable: costperDGA <dbl>
```

```
cost_IFs_sub %>% filter(food == "pf_redm") %>% head()
```

```
## # A tibble: 6 × 9
## # Rowwise:
##   subgroup food      food_type  mixed_dish wasted_coef_fndds wasted_prop_fndds inedible_c
##   <dbl> <chr>    <chr>      <chr>          <dbl>          <dbl>
## 1      0 pf_redm Grocery    Mixed          0.297          0.229
## 2      0 pf_redm Grocery    Non-Mixed      0.346          0.257
## 3      0 pf_redm Non-Grocery Mixed          0.285          0.222
## 4      0 pf_redm Non-Grocery Non-Mixed      0.330          0.248
## 5      0 pf_redm Total      Mixed          0.292          0.226
## 6      0 pf_redm Total      Non-Mixed      0.340          0.254
## # i 1 more variable: costperDGA <dbl>
```


6.6 Export Data

```
# cost IFs  
write_csv(cost_IFs_sub, paste("data_inputs/IMPACT_FACTORS/output_data/Impacts_cost  
  
# enviro IFs  
write_csv(enviro_IFs_sub, paste("data_inputs/IMPACT_FACTORS/output_data/Impacts_er
```

